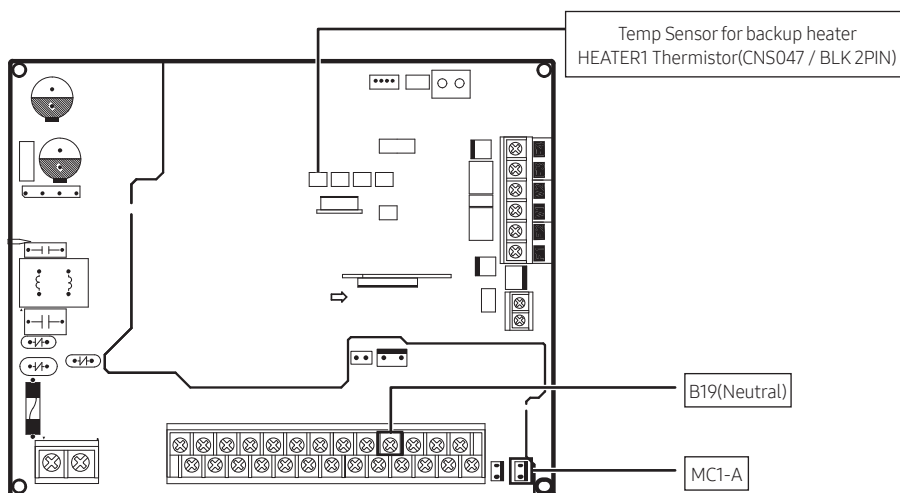


Backup heater



Connecting a external backup heater (MHC-300FP)

1. Connect a MHC-300FP with CNP001 and B19(Neutral)
2. Connect a Thermistor to CNS047(BLK Connector)
 - ▶ The N phase of the heater Signal can be connected to B5, B7, B11, B15 instead of B19 (Maximum 2 wires can be connected 1 Screw)



NOTE

- For the external Back-up heater connection, Only Samsung Heater kit(MHC-300FP) is compatible.
- It's just for providing a ON/OFF control signal.

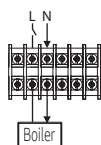
Connection of the back-up boiler

- ▶ This function is to determine which heating source can/will provide the space heating, either the heat pump system or the back-up boiler.
- ▶ To control the back-up boiler, configure FSV #4031~ #4033. Please refer to the controller manual for this setting.

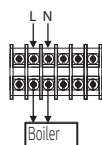
Description	No. of wires	Max. current	Thickness	Supply Scope
Back-up Boiler	2	50 mA	0.75mm ² H05RN-F or H07RN-F	Field supply (220-240 Vac, Output)



When it set back up
boiler on the control
kit (relay off)



When it order to back
up boiler operates
(relay on)

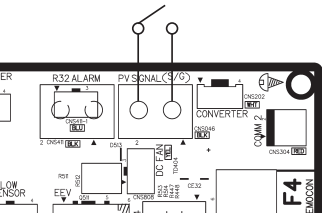


1. Before the installation, control kit should be turned off.
 2. Using the appropriate equipment to correct position of terminal block as shown on the diagram.
- * Heat pump does not work when the Back-up boiler operates.

Electrical installation

Connection of the Peak Power Control or PV control(Photovoltaics control)

Description	No. of wires	Max. current	Thickness	Supply Scope
Peak Power Control or PV control (Photovoltaics control)	2	-	-	Field supply



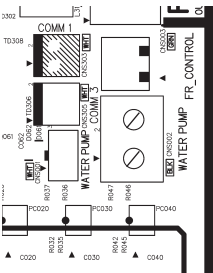
- * Function 1 (Peak Power Control)
 - ▶ This is a function that allows you to disable the booster heater, backup heater, and compressor operation of the outdoor unit depending on the power input contact.
 - ▶ If users make contracts with local electric power company for limiting the amount of power consumption when a surge in power usage, users can set FSV of "Forced off".
 - ▶ To control the Power Peak , configure FSV #5041~ #5043. Please refer to the controller manual for this setting.
- * Function 2 (PV[Photovoltaics] control)
 - ▶ This is for energy saving by using the solar energy.
 - ▶ To control PV, configure FSV #5081~ #5083. Please refer to the controller manual for this setting.
 - ▶ Cooling/heating operation: Enables the FSV value that is set only when the PV signal is in out mode.
 - Hot water operation: Activate immediately with the set FSV value

NOTE

- It operates according to the setting of FSV, and both functions can not be used at the same time. (PV Control / Peak power control)
- Except for domestic hot water mode, this function is only activated for the outing mode.

Connection of the the FR Control(Frequency ratio control)

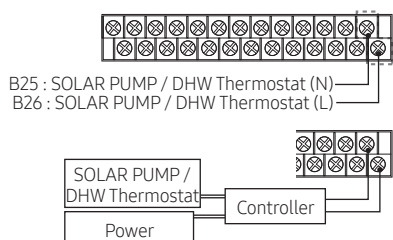
Description	No. of wires	Max. current	Thickness	Supply Scope
FR Control	2	-	-	Field supply



- ▶ The FR control function application is to limit the maximum frequency of the outdoor unit compressor. (if FSV #5051 = 1 "use")
- ▶ Method 1 : External DC signal Control uses a DC voltage of 0 ~ 10V (0v = 50%, ~10v = 150%)
- ▶ Method 2 : Demand ratio (DR) control through Modbus communication

Connection of the the Solar Pump / DHW thermostat

Description	No. of wires	Max. current	Thickness	Supply Scope
Solar pump / DHW Thermostat	2	22mA	0.75mm ² H05RN-F or H07RN-F	Field supply (220-240 Vac, output)



1. Before connecting the external control kit, make sure it is turned off.
2. Using the appropriate equipment to correct position of terminal block as shown on the diagram.
3. The external control kit must provide an output signal when Solar pump/ DHW Thermostat is operating.
4. It is the installers responsibility to connect the output of the control kit to the Solar Pump/ DHW Thermostat input terminal (B25-26). In operating mode, signal shall be around 230VAC (N-L). In non-operating mode, signal shall be around 0VAC (N-L).

When solar pump signal is On, Control kit DHW mode will be turned off.

If a solar pump for DHW is used, the signal input line from the solar pump can be connected as shown above

- if a solar pump is used, FSV#3061 should be set to 1

If DHW thermostat is used, the signal input line from the DHW thermostat can be connected as shown above

- if DHW thermostat is used, FSV#3061 should be set to 2

NOTE

- Solar pump and DHW thermostat cannot be used simultaneously.

Electrical installation

Connection guide of additional pump

Ensuring proper water flow is an important factor for performance and efficiency.

If the water flow is not adequate, consider installing additional pumps.

When installing an additional pump, the system's maximum allowable water flow rate and maximum water pressure should not be exceeded.

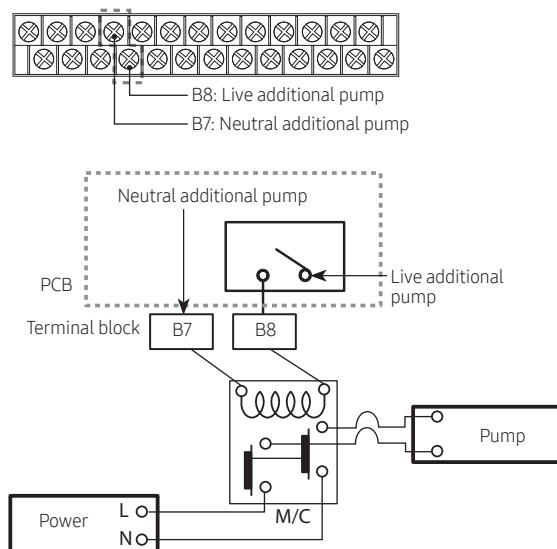
⚠ CAUTION

- Each terminal set (Live+Neutral) for additional pumps can supply ampere up to Max 1.0A.
- The maximum number of pumps that can be powered by the terminal block of the control kit is two. Therefore, if necessary, be sure to connect pumps other than the two a separate power source.

Case 1) AC pump

The terminal set (B8+B7) can supply ampere up to Max 1.0A. This model has default INV. PUMP on B7+B8, if you want to add additional Pump, please connect it to a separate power source. Please, Use this port as ON/OFF Control Signal.

Power supply (Pump)

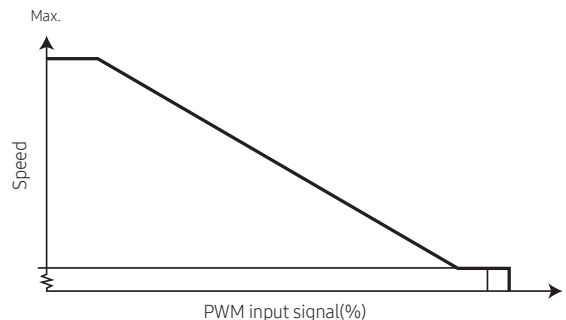


⚠ CAUTION

- The maximum allowable current that this terminal block can supply for the additional water pump is 50 mA.

Case 2) Inverter pump

Power supply connection is the same as the case1) AC pump
PWM characteristic curve

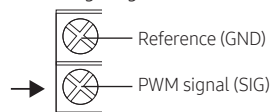


The additional pump should be the same type of product as the above graph.

Recommendation

Connect the PWM control line to the main control PBA output(CNS002) signal function for PWM control.

(Note: Wiring diagram)



GRUNDFOS UPMM 25-95 (Heating Type), SHINHOO GPA25-9H (Heating Type)

⚠ CAUTION

- If there is wrong wiring between PWM and reference, INV. Water Pump may not work or wrong operation.

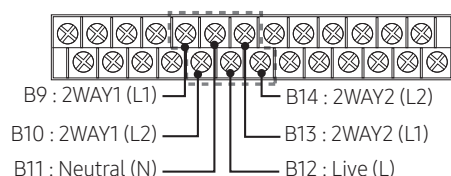
Connection of the 2-way valve (2zone control is not used as FSV Setting #4061="0")

1. 2Way Valve performs open/close operations according to cooling/heating Thermo On/Off control.
2. When setting up two thermostats, then;
 - 2way valve 1 operates according to the input signal of thermostat 1 and 2way valve 2 according to that of thermostat 2.
 - If you use the 2way valve output as the output of the Room Water Pump with buffer tank for Thermostat 1 and 2, please connect it to terminal block "normal close" and set FSV Setting #6041 to "0". [Reference: EHS CONTROL KIT Manual]

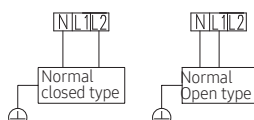
Description	No. of wires	Max. current	Thickness	Supply Scope
2 Way Valve	2+ground	50 mA	> 0.75 mm ² , H05RN-F or H07RH-F	Field supply (220-240 Vac, Output)

⚠ CAUTION

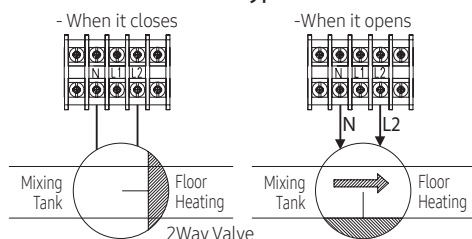
- When FSV#4041=0 is set without using the mixing valve, if the water outlet temperature is less than 16 degrees, 2way valve 1 is closed to prevent floor condensation.
 - Connect the output of 2way valve 1 for underfloor cooling shutoff.
- ▶ Connect the output of 2way valve 1 for underfloor cooling shutoff.



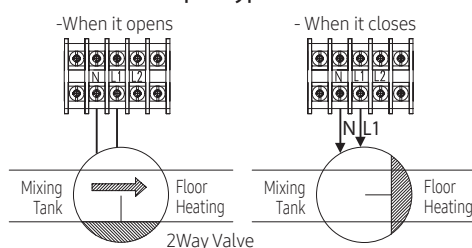
* Connection of 2 wires 2-way valve



In case of normal closed type



In case of normal open type



2-way motorized valve

- ▶ 220-240 Vac
 - ▶ 2 wires(Normal Open or Normal Close)
1. Before the installation, control kit should be turned off.
 2. Using the appropriate equipment to correct position of terminal block as shown on the diagram.
 3. Make sure what type is you use.
 - Normal OPEN or Normal CLOSED.

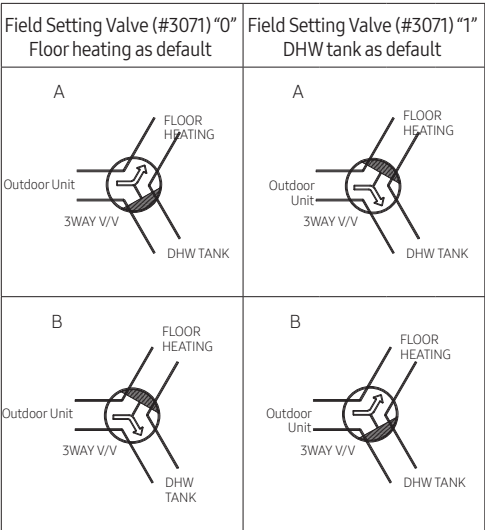
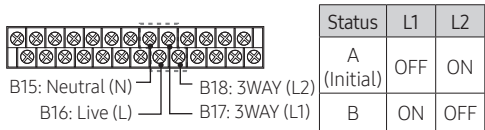
⚠ CAUTION

- There are 2 types of 2-way valve, normal open type and normal closed type. Make sure to connect terminals to right positions of terminal block. As detailed on the wiring diagram and illustrations above.

Electrical installation

Connection of the 3-way valve

Description	No. of wires	Max. current	Thickness	Supply Scope
Diverting type 3way valve	4	50mA	> 0.75 mm ² , H05RN-F or H07RN-F	Field supply (220-240 Vac, Output)



3-way diverting valve for water tank

- Diverting type cooling mode, UFH loops will be closed.
- 220-240 Vac

1. Before the installation, control kit should be turned off.
2. Using the appropriate equipment to correct position of terminal block as shown on the diagram.
3. Make sure what type of 3 way V/V you use.

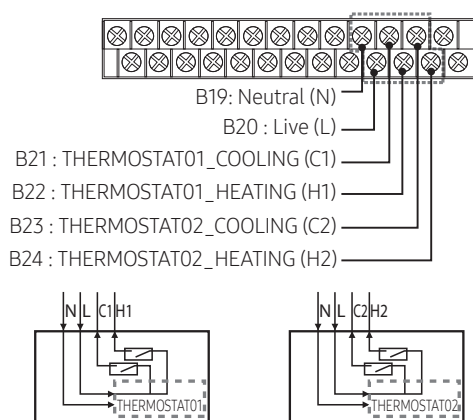
Connection of the thermostat

To use thermostat control, configure FSV #2091/#2092 to "Use." Please refer to the controller manual for this setting. Cooling/Heating operation and water temperature are determined by the Thermostat input signal and Water Law setting. Please refer to the controller manual for this setting.

NOTE

- For thermostat control, FSV Setting #4061 must be set to "0".

Description	No. of wires	Max. current	Thickness	Supply Scope
Room Thermostat	4	#19, #20 : 50 mA (Output, Thermostat Power) #21, #22, #23, #24 : 22 mA each Pin (Input)	> 0.75 mm ² , H05RN-F or H07RH-F	Field supply (220-240 Vac, Output)



- Before the installation, control kit should be turned off.
- Using the appropriate equipment to correct position of terminal block as shown on the diagram.
- Make sure what type is you use.
 - If the operating mode (heating/cooling) of thermostat 1 and thermostat 2 are different, the operating mode is determined based on thermostat 2.

CAUTION

- Product will not operate when signal for cooling and heating mode is inputted at the same time.

When setting up two thermostats;

- For heating operation, please install Thermostat 01 to a room that requires lower water temperature (ex, underfloor heating) and Thermostat 02 to a room that requires higher water temperature (ex, radiators).
- For cooling operation, please install Thermostat 02 to a room that requires lower water temperature (ex, radiators) and Thermostat 01 to a room that requires higher water temperature (ex, underfloor heating).
- In order to properly control the two thermostats with different temperature settings, it is necessary to connect the mixing valve and the mixing temperature sensor (TW4) [Optional parts].
- When controlling the thermostat in a multi-room (Two thermostat control), please refer to the "Connection of the 2-way valve" wiring works for valve control.
- Please find additional installation information refer "Connection of the 2-way valve" [Page 31] and "Connection of the mixing valve" [Page 31]

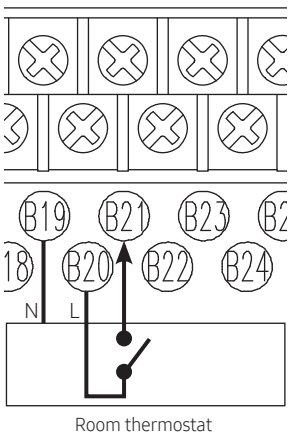
CAUTION

- When using thermostat 01 in cooling mode, to prevent floor condensation, if the room water temperature is below 16 °C, the 2-way valve of thermostat 01 is automatically closed.

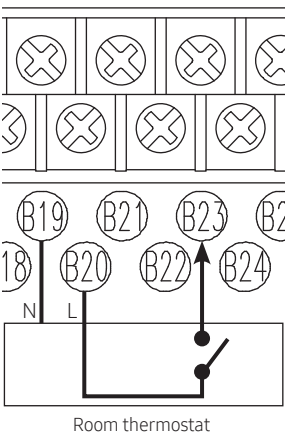
Electrical installation

Example

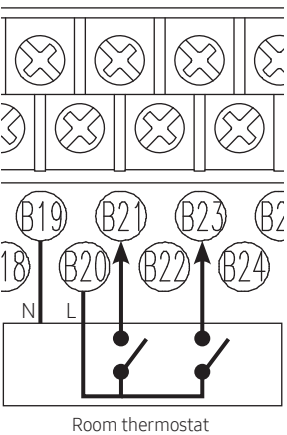
Thermostat01 only : cooling mode



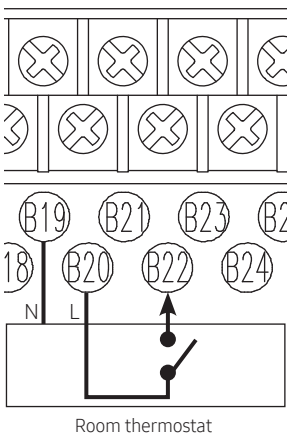
Thermostat02 only : cooling mode



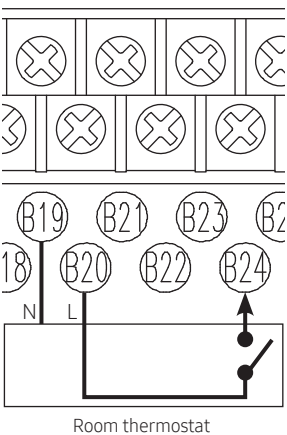
Thermostat01, Thermostat02 :
cooling mode



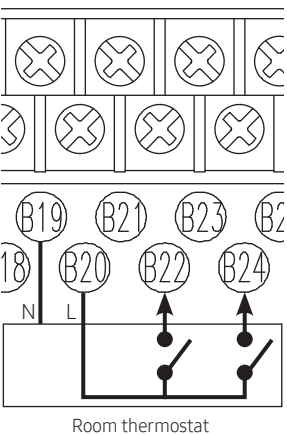
Thermostat01 only : heating mode



Thermostat02 only : heating mode



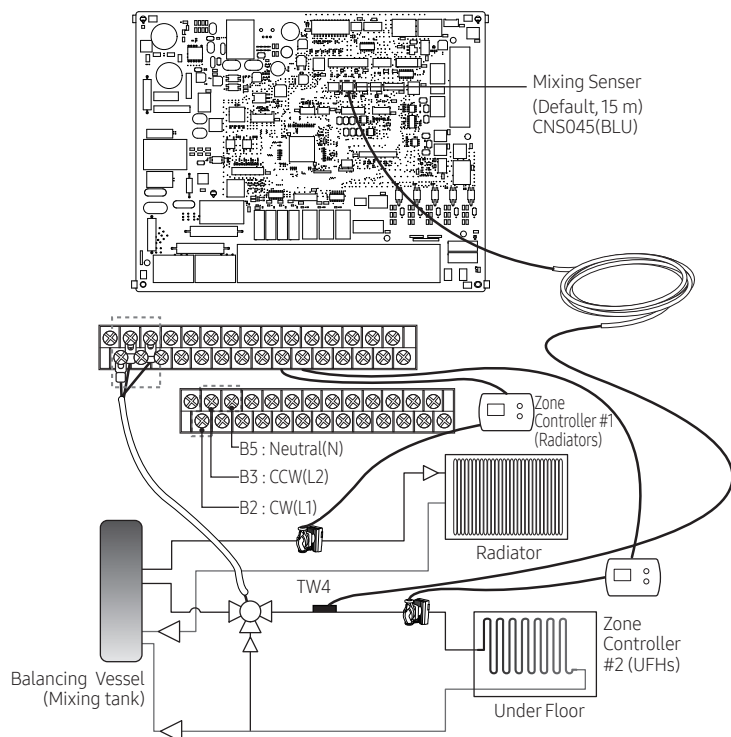
Thermostat01, Thermostat02 :
heating mode



⚠ WARNING

- Before completing installation of Room thermostat, check the wiring method in a manual of Room thermostat to output L line.

2 Zone with Two thermostat control



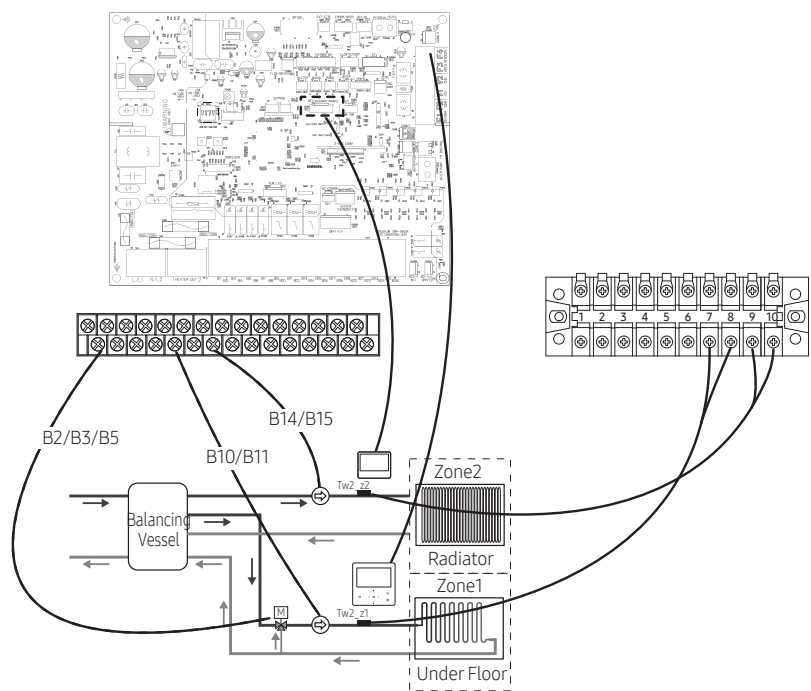
1. Before the installation, control kit should be turned off.
2. Using the appropriate equipment to correct position of terminal block as shown on the diagram.
3. Install the supplied Mixing temperature sensor (TW4) on the rear part of the mixing valve. Install TW4 Sensor within 1m of Mixing Valve.
4. Set the FSV value #4041~ #4046. Please refer to the controller manual for this setting.

CAUTION

- When using Zone control (FSV 4061 = 1), ignore Thermostat signal.

Electrical installation

2- Zone Control [FSV #4061 =1]



You can operate the 2-zone control using a mixing value, water-out temperature sensors, and built-in or external room temperature sensors installed in a wired remote controller.

When both zones are simultaneously Thermo on, the operation is performed based on Zone2. Therefore, set the zone that you want to have the higher set temperature to Zone2.

(The mixing valve must be installed in the zone that you want to have the lower set temperature.)

1. Install the mixing valve. (See "Installation of mixing valve.")
2. Install the water-out temperature sensors (Tw2_z1, Tw2_z2) for all zones.
3. Unlike the zone control with a thermostat, connect the water pump signal lines to the product.
 - Zone1 water pump connection: B10 (L1) + B11 (N)
 - Zone2 water pump connection: B14 (L1) + B15 (N)
4. FSV 4061 = 1: Enable the 2-zone control using the wired remote controller.

- * If you want to operate the 2-zone control by using water-out temperatures, you have only to complete steps 1 to 4 above.
- * If you want to operate the 2-zone control by using room temperatures and built-in temperature sensors in wired remote controllers, you must install two wired remote controllers in each room.

(If you use external room temperature sensors, you can control each room temperature with only one wired remote controller.)
Select a mixing valve from the manufacturers as below (recommended)

Maker		BELIMO	SIEMENS	HONEYWELL
Model code	3 Way Valve	R3020-6P3-S2	VXP45.20-4 (kvs 4)	V5011E1213
	Actuator	LR230A(-S)	SSB31	ML6420A3015
Running time		90 sec.	150 sec.	60 sec.
FSV(#4046) setting		9	15	6

Specification of power cable

• 1 Phase

- The power cables are not supplied with the air to water heat pump.
- Power supply cords of parts of appliances for outdoor use shall not be thinner than polychloroprene sheathed flexible cord (Code designation IEC:60245 IEC 57 / CENELEC:H05RN-F)
- This equipment complies with IEC 61000-3-12.

Outdoor unit	Rated		Voltage Range		MCA	MFA
	Hz	Volts	Min	Max	Min Circuit Amps.	Max Fuse Amps.
AE050CXYBEK	50	220-240	198	264	16.1	17.6
AE080CXYBEK	50	220-240	198	264	26	28.6
AE120CXYBEK	50	220-240	198	264	32	35.2
AE160CXYBEK	50	220-240	198	264	32	35.2

• 3 Phase

- The power cables are not supplied with the air to water heat pump.
- Power supply cords of parts of appliances for outdoor use shall not be lighter than polychloroprene sheathed flexible cord (Code designation IEC:60245 IEC 66 / CENELEC:H07RN-F)
- This equipment complies with IEC 61000-3-12 provided that the short-circuit power (SSC) is greater than or equal to 3.3[MVA] at the interface point between the user's supply and the public system. It is the responsibility of the installer to ensure, by consultation with the energy company if necessary, that the equipment is connected only to a supply with a short-circuit power (SSC) greater than or equal to 3.3[MVA].

Outdoor unit	Rated		Voltage Range		MCA	MFA
	Hz	Volts	Min	Max	Min Circuit Amps.	Max Fuse Amps.
AE080CXYBGK	50	380-415	342	457	16.1	17.7
AE120CXYBGK	50	380-415	342	457	16.1	17.7
AE160CXYBGK	50	380-415	342	457	16.1	17.7

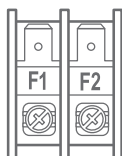
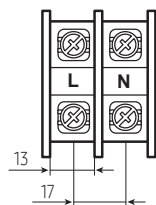
Terminal block specification

• 1 Phase

(Unit: mm)

AC power: M5 screw

Communication: not used

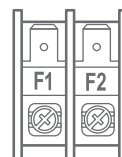
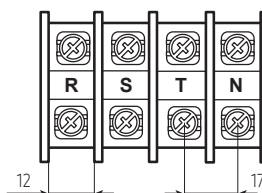


• 3 Phase

(Unit: mm)

AC power: M5 screw

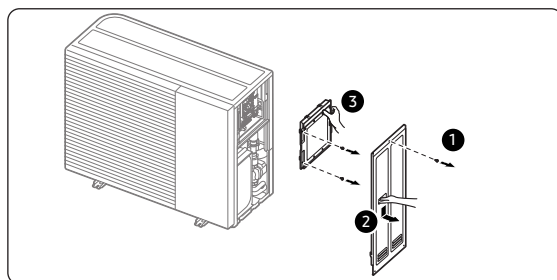
Communication: not used



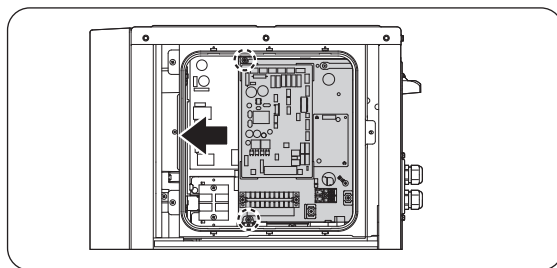
Tightening Torque (kgf • cm)		
M4	12~18	Communication : F1, F2
M5	20~30	1phase AC power : L, N
		3phase AC power : L1(R), L2(S), L3(T), N

Outdoor wiring

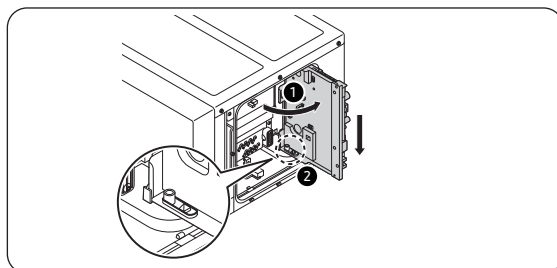
1 Open the switch side cover.



2 Remove 2 screws and pull the plate to the left.

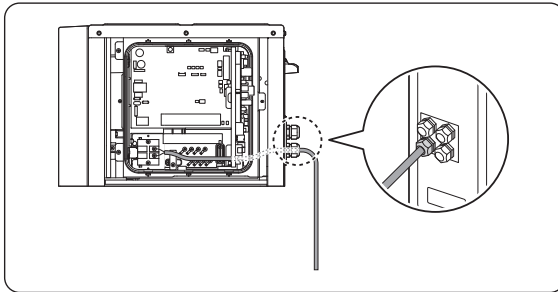


3 If the plate is rotated to where the hook is, it will be fixed to the hook.



Electrical installation

- 4 Insert the cables at the rear of the unit, and route them through the factory mounted cable sleeves into the switch box.



Power and communication cable configuration

- Power supply cable must be guided through the knock-out hole on the bottom-right or right side of the cabinet.
- Guide the communication cable through the designated knockout hole on the bottom-right side of the front part.
- Install the power and communication cable using a separate cable protection tubing.
- Fix the conduit to the knock-out hole on the outdoor unit by using a conduit bushing and nut.

To connect the power supply

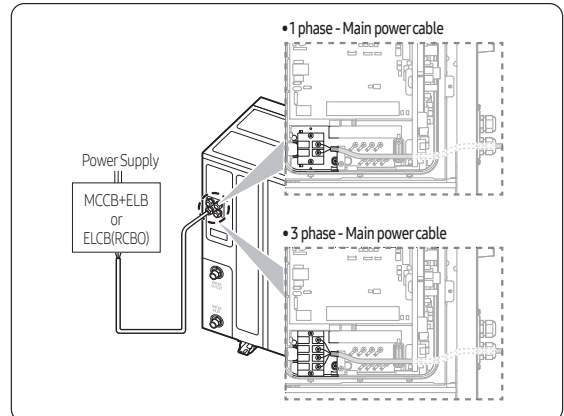
- Two cables must be connected to the outdoor unit
 - The communication cable between the indoor part and outdoor unit.
 - The power cable between the outdoor unit and the auxiliary breaker.
- Especially for the Russian and European market, before installation, the supply authority should be consulted to determine the supply system impedance to ensure compliance.

⚠ CAUTION

- You should connect the power cable into the power cable terminal and fasten it with a clamp.
- The unbalanced power must be maintained within 2% of the supply rating.
 - If the power is unbalanced greatly, it may shorten the life of the electrical components. If the unbalanced power is exceeding 4% of the supply rating, the control kit is protected, stopped and an error code is displayed.
- To protect the product from water and possible shock, keep the power cable and the connection cord of the control kit and outdoor units within ducts (with appropriate IP rating and material selection for your application).
- Ensure that the main supply connection is made through a disconnection switch within your arm's reach, that disconnects all poles, with contact gap of a least 3 mm.

Route the cable through the frame

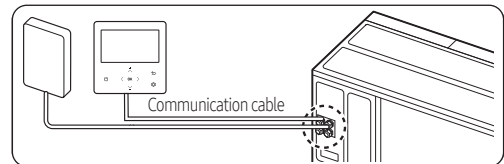
- Connect the wires to the terminal block and fix the cable with the cable tie.



- When installing electrical wiring: tension on the cable(s) must be avoided.
- Earth wire for the outdoor unit cables must be clamped to a suitable ring terminal clamp (not supplied)
- For the power cable, use the grade H07RN-F or H05RN-F materials.
- Power supply cords of parts of appliances for outdoor unit use shall not be thinner than polychloroprene sheathed flexible cord. (Code designation IEC:60245 IEC 57 / CENELEC: H05RN-F or IEC:60245 IEC 66 / CENELEC: H07RN-F)

To connect the communication cable

- The communication cable between the indoor part and outdoor unit.
- Route the cable through the frame.

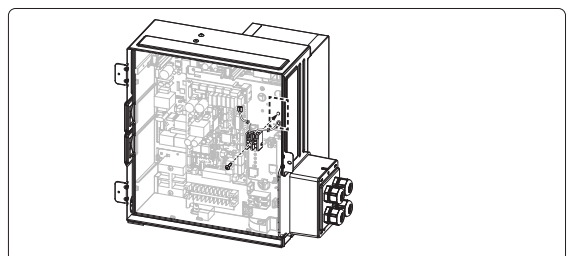


- Specification of communication cable

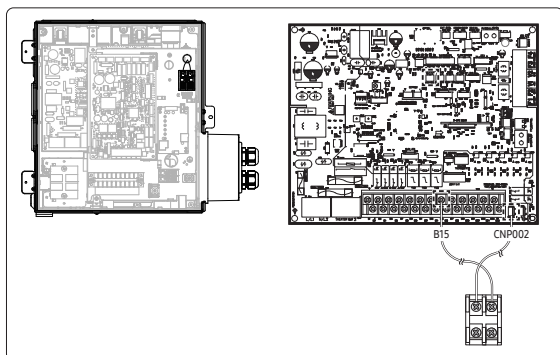
Communication cable
0.75 mm ² , 2 wires shielded

To connect the booster heater

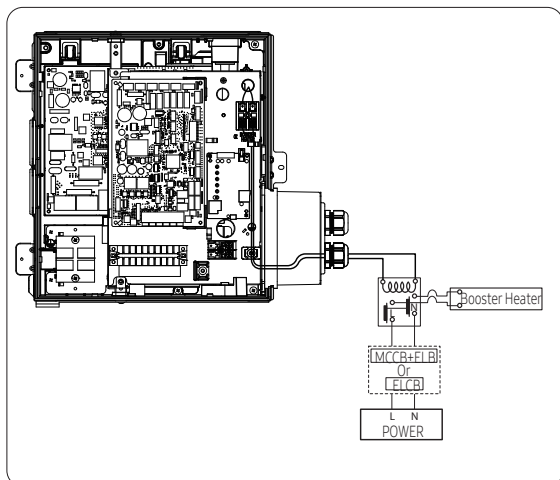
- Find Booster Heater Terminal Block KIT.
- Install the Terminal Block KIT after checking the hole for screw.



- 3 Connect the wire to B15 (Neutral) and CNP002 through the marked hole.
- * It's just for providing a ON/OFF Control Signal of the Booster Heater.
 - * The N Phase of the HEATER Signal can be connected to B5,B7,B11 instead of B15 (Maximum 2Wires can be connected 1 Screw)



- 4 Connect cable(field supply) between the Terminal Block and the Booster Heater.
- * Booster Heater Relay should be installed separately on a distribution board to prevent fire risk in case of refrigerant leakage.



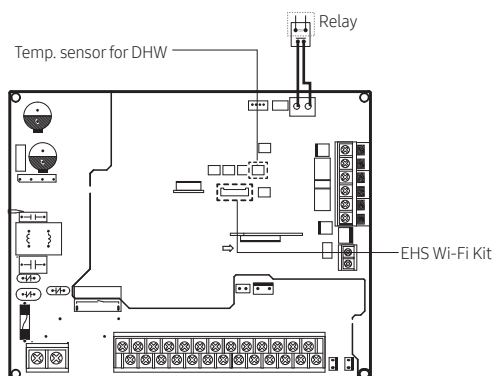
Description	No. of wires	Max. current	Thickness	Supply Scope
Booster Heater	2+Ground	50 mA	0.75 mm ² , H05RN-F or H07RH-F	Field supply (220-240 Vac, Output)

NOTE

- This port can NOT supply enough power for driving a Booster Heater.
- It's just for providing a ON/OFF control signal.

Temp. Sensor for DHW, PV / Peak power control Signal

External wiring to control a switch of relay by an installer



Connecting a temp. sensor wire into DHW

1. Put the sensor side of a temp. sensor wire into the designated location in a DHW.
2. Connect the other side of the line at CNS042.

Connecting a PV(Photovoltaic) Signal / Peak power control Signal

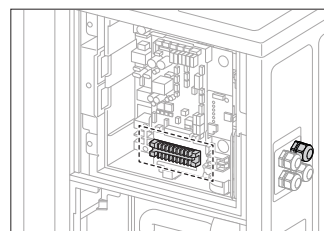
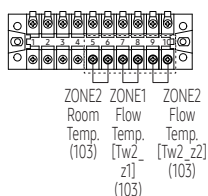
1. Install as above diagram.
2. Connect the PV / Peak power control signal wire to the 'CNS046' connector.

NOTE

- It operates according to the setting of FSV, and both functions can not be used at the same time. (PV Control / Peak power control)

To connect external sensors

- When the room temperature for zone2 is controlled via a remote sensor, please connect the zone sensor (white, supplied) to the correct terminals (15/16)
- To control the supply water temperature for both zone1 and zone2, it is required to install the supplied sensor(s) at the correct position. Please connect zone sensors (Tw2_z1 and Tw2_z2) to the below mentioned terminal block at the correct terminals (17/18 and 19/20)



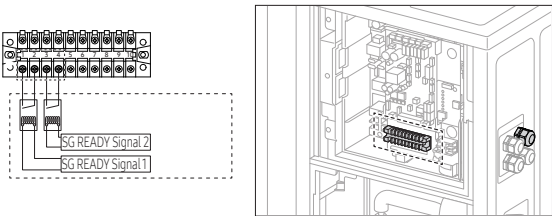
CAUTION

- When connecting sensors, use a Thermistor with the specifications of 10 kΩ at 25 °C, B constant = 3435 k.

Electrical installation

To connect smart grid

- To control the ClimateHub via external smart grid (SG) input signals (potential free, dry contact), please connect to the correct terminals (Signal1 (1/2), Signal2 (3/4)).
If needed use relays, not supplied with the product.



SG READY Signal 1	SG READY Signal 2	Description
Short	Open	Forced thermo off operation
Open	Open	Normal operation
Open	Short	Heating / DHW setting temperature 1step-up operation
Short	Short	Heating / DHW setting temperature 2step-up operation

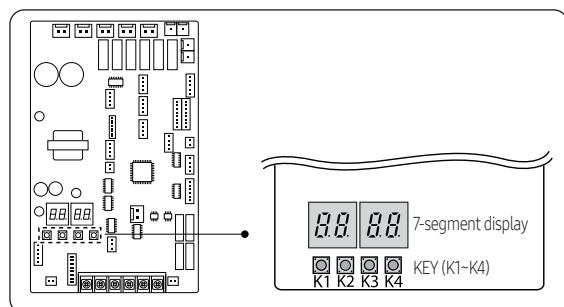
CAUTION

- These parts are optional and not included with the product.
- Turn off the **ELCB** first before connecting the **SG Ready**.

System configuration

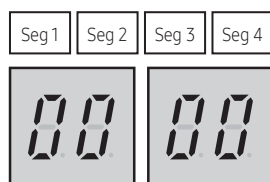
Setting the outdoor options via K-button tactile switches

Setting the correct options for the outdoor unit for site specific conditions via K-button tactile switches and explanation of their functions.



Setting the option

- Press and hold K2 to enter the option setting. (Only available when the operation is stopped)
 - When entering the option setting, the display will show the following:



- Seg 1 and Seg 2 will display the number for the selected option.
 - Seg 3 and Seg 4 will display the number for the set value of the selected option.
- After entering the option setting mode, shortly press the K1 switch to go to the desired option to change, indicated by Seg 1, Seg 2 and select the desired option. Example)



- After selecting the desired option, you can shortly press the K2 switch to adjust the value of the option, as indicated by Seg 3 and Seg 4, and change the function setting for the selected option.



- After selecting the function setting for the selected options, press and hold the K2 switch for 2 seconds to store the value. The changed value of the option will be saved when the entire segment display blinks and tracking mode begins.

CAUTION

- If the above instruction is not exactly followed, there is a risk that changed option settings are not saved.
- To restore the previous setting, press and hold the K1 button.
- To call for the factory default setting, press and hold K4 during the option setting mode.
 - After the factory default value is displayed, it needs to be saved. Press and hold the K2 button. When the segments show that tracking mode is in progress, the setting is saved.

Key Option

Option	SEG1	SEG2	SEG3	SEG4	Function of the option
Current restriction rate	0	0	0	0	100% (Factory default)
			0	1	95%
			0	2	90%
			0	3	85%
			0	4	80%
			0	5	75%
			0	6	70%
			0	7	65%
			0	8	60%
			0	9	55%
			1	0	50%
			1	1	No restriction
Defrost operation	0	1	0	0	Basic
			0	1	Option
Fan speed correction for outdoor unit	0	2	0	0	Basic
			0	1	Option
Quiet mode	0	3	0	0	Low noise (Basic)
			0	1	Level1
			0	2	Level2
			0	3	Level3
Channel address	0	4	A	U	Automatic setting (Factory default)
			0~15		Manual setting for channel 0~15
Snow accumulation prevention control	0	5	0	0	Enabled (Factory default)
			0	1	Disabled
Base Heater	0	6	0	0	Disabled
			0	1	Enabled (Factory default)
Operation mode	0	7	0	0	Heat pump (Factory default)
			0	1	Cooling only (Disabled)
			0	2	Heating only
Energy saving mode	0	8	0	0	Disabled (Factory default)
			0	1	Enabled

System configuration

⚠ WARNING

- Incorrect handling of the thermostat, safety valve or other valves may lead to tank rupture or plate heat exchanger damage. When servicing the unit follow the instructions carefully:
 - Always turn off the main power supply when the water supply is being shut off.
 - Test the free operation of the safety valve regularly by opening the valve ensuring the water flows freely.
 - Electrical connection and all servicing of the electrical components should only be carried out by an authorized electrician.
 - Fitting and all servicing of plumbing fixtures should only be carried out by an authorized installer.
 - When replacing the thermostat, safety valve, or any other valve or part supplied with this unit, use only approved parts of the same specification.

Testing operations

- Check the power supply between the outdoor unit and the auxiliary circuit breaker.
 - 1 phase power supply: L, N
 - 3 phases power supply: R, S, T, N
- Check the control panel.
 - Check that you have connected the power and communication cables correctly. (If the power cable and communication cables are mixed up or connected incorrectly, the PCB will be damaged.)
 - Check the temp. sensor, drain pump/hose, and display are connected correctly.
- Press K1 or K2 on the outdoor unit PCB to run the test mode and stop.

K1 (Number of Press)	KEY operation	7-segment display
1 time	Trial operation in Heating mode	"K" "1" "BLANK" "BLANK"
2 times	Vacuuming (Outdoor unit address 1)	"K" "2" "BLANK" "1"
3 times	Inverter Fault Detection (Comp #1)	"K" "3" "1" "1"
4 times	End Key operation	-

K2 (Number of Press)	KEY operation	7-segment display
1 time	Trial operation in Cooling mode	"K" "4" "BLANK" "BLANK"
2 times	Discharge mode of DC link voltage	"K" "5" "0" "k"
3 times	Forced defrost operation	"K" "6" "BLANK" "BLANK"
4 times	Inverter compressor 1 check	"K" "7" "BLANK" "BLANK"
5 times	End Key operation	-

K3 (Number of Press)	KEY operation	7-segment display
1 time	Initialize (Reset) setting	Same as initial state

- View Mode: When the K4 switch is pressed, you can see information about our system state as below.

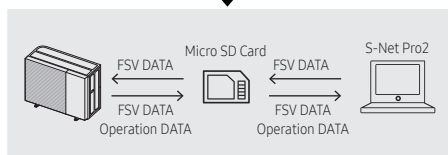
K4 (Number of press)	KEY operation	Display on segment	
		SEG 1	SEG 2,3,4
1 time	Capacity of outdoor unit	1	16 HP → 0,1,6
2 times	Order frequency of the compressor	2	120 Hz → 1,2,0
3 times	High pressure (kg/cm ²)	3	15.2 K → 152
4 times	Low pressure (kg/cm ²)	4	4.3 K → 043
5 times	Discharge temperature Compressor	5	87 °C → 087
6 times	IPM temperature	6	87 °C → 087
7 times	CT sensor value	7	2 A → 020
8 times	Suction temperature	8	-42 °C → -42
9 times	COND OUT temperature	9	-42 °C → -42
10 times	EVA in temperature	A	87 °C → 087
11 times	Compressor top temperature	B	87 °C → 087
12 times	Outdoor temperature	C	-42 °C → -42
13 times	EVI inlet temperature	D	-42 °C → -42
14 times	EVI outlet temperature	E	-42 °C → -42
15 times	Main EEV step	F	2000 → 200
16 times	EVI EEV step	G	300 → 300
17 times	Fan step (ssr or bldc)	H	13 Step → 0, 1, 3
18 times	Current frequency Compressor	I	120 Hz → 1, 2, 0
19 times	EVI SOL EEV step	J	300 → 300
20 times	Inverter pump output	K	100 % → 100

K4 (Number of press) Press and hold the K4 to enter the setting	Displayed content	Display on segment	
		Page 1	Page 2
1 time	Main version	MAIN	Version (ex. 1412)
2 times	Inverter version	INV	Version (ex. 1412)
3 times	EEP version	EEP	Version (ex. 1412)
4 times	Automatically assigned address of the units	AUTO	SEG1,2 Indoor unit: "A", "0" MCU: "C", "1" Address (ex:07)
5 times	Manually assigned address of the units	MANU	SEG1,2 Indoor unit: "A", "0" Address (ex:15)

FSV Upload / Download

To make an easy installation and service possible, please use an SD card and the wired remote controller's function.

Field Setting Value		
10**	Remote Controller	>
20**	Water Law	>
30**	DHW	>
40**	Heating	>
50**	Others	>
Simple Setting		
FSV Upload / Download		



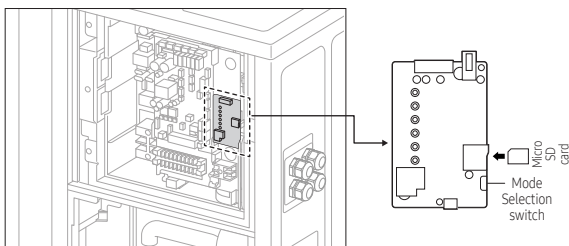
How to upload or download the Field Setting Values (example)

- 1 Insert an SD card into the Sub PBA SD Card slot on the pump integrated outdoor unit.
- 2 Select Field Setting Value in the Service mode.
- 3 Press the $\wedge$ or $\vee$ button to select FSV Upload / Download.

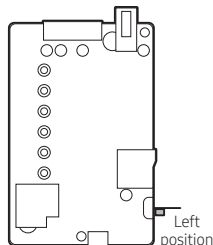


NOTE

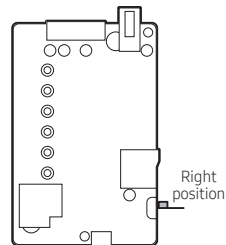
- Upload: Uploads the FSV data of the pump integrated outdoor unit to the SD card.
 - Download: Downloads the FSV data of the SD card to the pump integrated outdoor unit.
- 1 Download the wired remote controller image, change the file name to "IMAGE.BIN," and then download to microSD Card.
 - 2 Download the wired remote controller program, change the file name to "MICOM.BIN," and then download to microSD Card.
 - 3 Insert the microSD card with the wired remote controller active, and then reset the system. For system reset, press the $\leftarrow$ and $\otimes$ buttons at the same time for more than 5 seconds.
 - 4 The download to microSD card is performed as follows:
 - Download proceeds in the order of first IMAGE and then MICOM.
 - When the update fails, the wired remote controller is reset automatically and the update proceeds again.
 - 5 When the download is finished, tracking starts automatically. When tracking is completed, remove the microSD Card.
- Add the function of FSV download / upload and data backup with SD card.
 - When FSV is download / upload, using the remote controller.



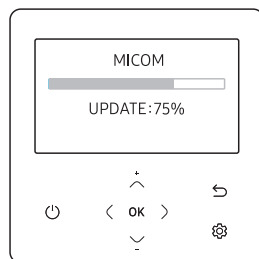
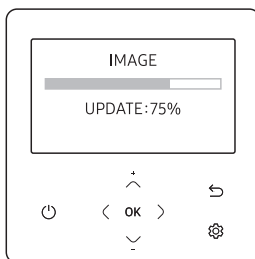
Left position: Field Setting Value (FSV) mode



Right position: Operating Data Backup mode



- Micro SD card is a separate purchase, using 8 ~ 32GB is possible.



CAUTION

- Be sure to use the microSD Card after formatting in FAT16 or FAT32.
- The microSD Card supports the SD or SDHC with the capacity of 1 GB to 32 GB.
- The update is performed only when the file version in the microSD Card differs from the one of the Wired Remote Controller.
- When the screen is frozen for more than 3 minutes after completing the microSD card update by 100%, product inspection is required.
- Delete the files except for the ones for download. (Only IMAGE.BIN and MICOM.BIN are required.)

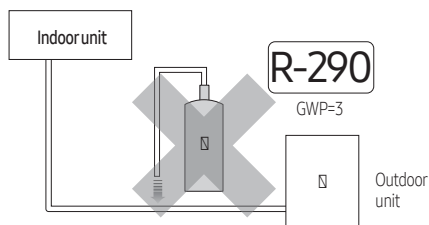
Charging refrigerant

- Measure the quantity of the refrigerant according to the length of the liquid side pipe. Due to the small quantity of the R-290 charge, an accurate weighing of the R-290 charged in the system is mandatory to ensure proper performance.

Important information: regulation regarding the refrigerant used

Please fill in the following with indelible ink on the refrigerant charge label supplied with this product and on this manual.

- ☐ the factory refrigerant charge of the product.
- ☐ the additional refrigerant amount charged in the field.



Unit	kg	tCO ₂ e
☐ a		
☐ b	DO NOT CHARGE	

Refrigerant type	GWP value
R-290	3

- GWP: Global Warming Potential
- Calculating tCO₂e: kg x GWP/1000

NOTE

- a Factory refrigerant charge of the product: see unit name plate.
- b Additional refrigerant amount charged in the field. (Refer to the above information for the quantity of refrigerant replenishment.)

CAUTION

- The filled-out label must be adhered in the proximity of the product charging port.
(ex. onto the inside of the stop valve cover.)

Troubleshooting

Troubleshooting of error code

Display	Explanation	Error Source
E108	Setting Address Duplicate Error	OUTDOOR UNIT/ Hydro Unit
E120	Zone2 Indoor Room Temperature Sensor Short/Open	Hydro Unit
E121	Error of ROOM Temperature Sensor in Indoor unit short/open	Hydro Unit
E122	Error of Evaporator_in Sensor of Indoor unit short/open	Hydro Unit
E123	Error of Evaporator_out Sensor of Indoor unit short/open	Hydro Unit
E162	Outdoor UNIT EEPROM Error	OUTDOOR UNIT
E163	EEPROM OPTION SETTING Error	OUTDOOR UNIT
E177	In hydro box, take place emergency signal Error	Hydro Unit
E201	CONTROL KIT / Outdoor UNIT Communication Error (Matching Error)	OUTDOOR UNIT
E202	CONTROL KIT / Outdoor UNIT Communication Error	OUTDOOR UNIT
E205	Communication Error Between Outdoor Unit Inv Micom - Fan Motor Micom	OUTDOOR UNIT
E221	Outdoor Temperature Sensor Error (open/short)	OUTDOOR UNIT
E231	COND_OUT Main Temperature Sensor Error (open/short)	OUTDOOR UNIT
E241	COND OUT Sensor of Outdoor Unit breakaway Error	OUTDOOR UNIT
E251	Discharge Temperature Sensor Error (open/short)	OUTDOOR UNIT
E262	Discharge Sensor breakaway Error	OUTDOOR UNIT
E266	Comp Top Sensor breakaway Error	OUTDOOR UNIT
E269	SUCTION Sensor breakaway Error	OUTDOOR UNIT
E276	Compressor Top Temperature Sensor Error (open/short)	OUTDOOR UNIT
E291	High Pressure Sensor Error (open/short)	OUTDOOR UNIT
E296	Low Pressure Sensor Error (open/short)	OUTDOOR UNIT
E308	Suction Sensor Error (open/short)	OUTDOOR UNIT
E321	EVI Inlet Sensor Error (open/short)	OUTDOOR UNIT
E322	EVI Outlet Sensor Error (open/short)	OUTDOOR UNIT
E381	Outdoor Inverter1 Controller PCB Overheat	OUTDOOR UNIT
E403	Protect for Freezing Control Error	OUTDOOR UNIT
E407	COMP down due to High PressureSensor Protection Control	OUTDOOR UNIT
E410	COMP down due to Low PressureSensor Protection Control	OUTDOOR UNIT
E416	Comp down due to discharge Temperature	OUTDOOR UNIT
E425	Outdoor Reverse Phase or Missing Phase Detect Error	OUTDOOR UNIT
E428	COMP down by Compression Ratio control Error	OUTDOOR UNIT
E436	Protect for Freezing Burst Control Error	OUTDOOR UNIT
E438	EVI EEV Opening Error	OUTDOOR UNIT

Display	Explanation	Error Source
E439	Refrigerant Leakage Error (Detect when the system is not operated)	OUTDOOR UNIT
E440	Forbid Heat mode Operation when Outdoor Temperature is over 43 °C	OUTDOOR UNIT
E441	Forbid Cooling Mode when Outdoor Temperature is Low 10 °C	OUTDOOR UNIT
E443	No startup due to Low pressure	OUTDOOR UNIT
E458	Outdoor unit Fan Error	OUTDOOR UNIT
E461	[Inverter] COMP Operating failure	OUTDOOR UNIT
E462	All currency control COMP Stop Or CT2 Low currency	OUTDOOR UNIT
E464	[Inverter] DC peak Error	OUTDOOR UNIT
E465	Compressor V limit Error	OUTDOOR UNIT
E466	[Inverter] DC-Link voltage under/over Error	OUTDOOR UNIT
E467	COMP Revolute Error	OUTDOOR UNIT
E468	[Inverter] Comp Current Sensor Error	OUTDOOR UNIT
E469	DC Link Sensor Error	OUTDOOR UNIT
E471	[Inverter] OTP Error → Outdoor unit EEPROM Read/Write Error (OTP error)	OUTDOOR UNIT
E474	[Inverter] IPM Heat Sink Error	OUTDOOR UNIT
E475	Outdoor unit BLDC Fan Error	OUTDOOR UNIT
E483	H/W DC_link over voltage Error	OUTDOOR UNIT
E484	PFC Overload Error	OUTDOOR UNIT
E485	[Inverter] Input Current Sensor Error (open/short)	OUTDOOR UNIT
E488	AC Input Voltage Sensor Error	OUTDOOR UNIT
E500	IPM Overheat Error for Inverter COMP	OUTDOOR UNIT
E507	Comp Down due to high pressure or High Pressure Switch Open	OUTDOOR UNIT
E536	PHE refrigerant leakage error	OUTDOOR UNIT
E554	Refrigerant leakage error (detection during system operation)	OUTDOOR UNIT
E563	INDOOR UNIT Mixed Install Error	OUTDOOR UNIT
E590	[Inverter] Data flash Error	OUTDOOR UNIT
E897	Water Tank in sensor error(Short/Open)	Indoor unit
E899	Zone1 Tw Temperature Sensor Short/Open	Hydro Unit
E900	Zone2 Tw Temperature Sensor Short/Open	Hydro Unit
E901	Water Inlet Sensor Error (open/short)	OUTDOOR UNIT
E902	Water Outlet Sensor Error (open/short)	OUTDOOR UNIT
E904	Water TANK Sensor SHORT / OPEN	Hydro Unit
E906	Outdoor EVA in Sensor Error (open/short)	OUTDOOR UNIT
E907	Error due to pipe rupture protection	CONTROL KIT
E908	Error due to freeze prevention(Re-Operation is possible)	CONTROL KIT
E909	Error due to freeze prevention(Re-Operation is impossible)	CONTROL KIT
E910	Water Temperature Sensor on water Outlet pipe is detached	Hydro Unit
E911	Flow Swtich Open Error	Hydro Unit

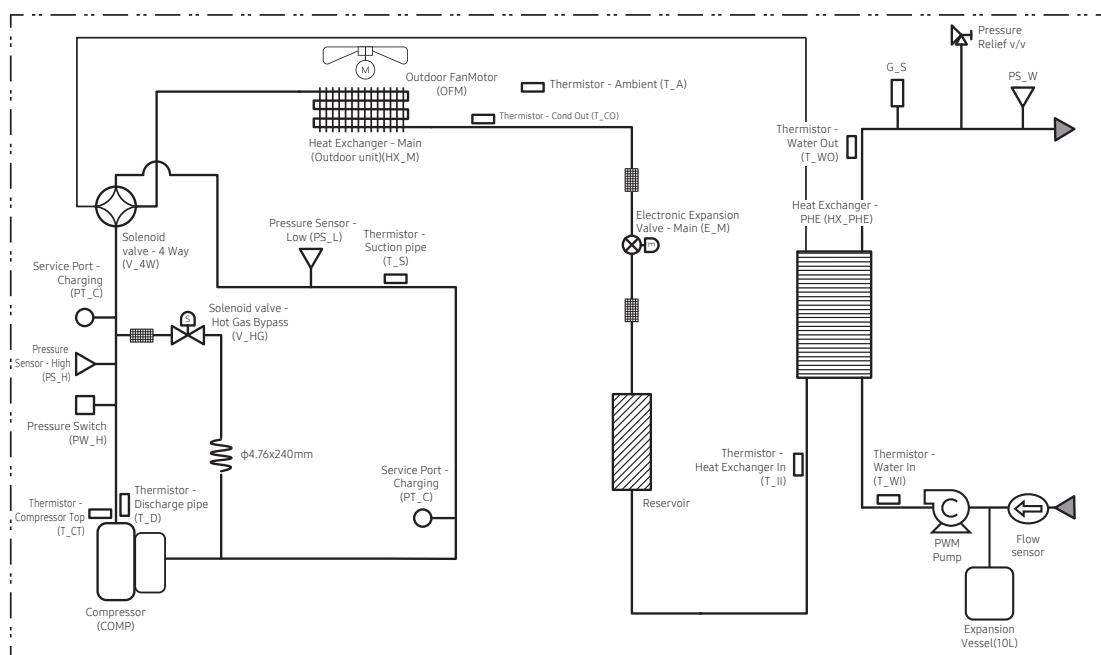
Troubleshooting

Display	Explanation	Error Source
E912	Flow Switch Close Error	Hydro Unit
E913	Six times detection for Flow Switch Error(Re-Operation is not possible)	CONTROL KIT
E914	Error due to Incorrect Thermostat Connection	Hydro Unit
E915	Error on DC fan(Non-operating)	CONTROL KIT
E916	Mixing Sensor Short/Open	Hydro Unit
E919	Disinfection Operation Incomplete Error	Hydro Unit
E973	Water pressure sensor research (Short/Open)	OUTDOOR UNIT

Technical data

Refrigerant circuit diagram

- AE050CXYB**/AE080CXYB**

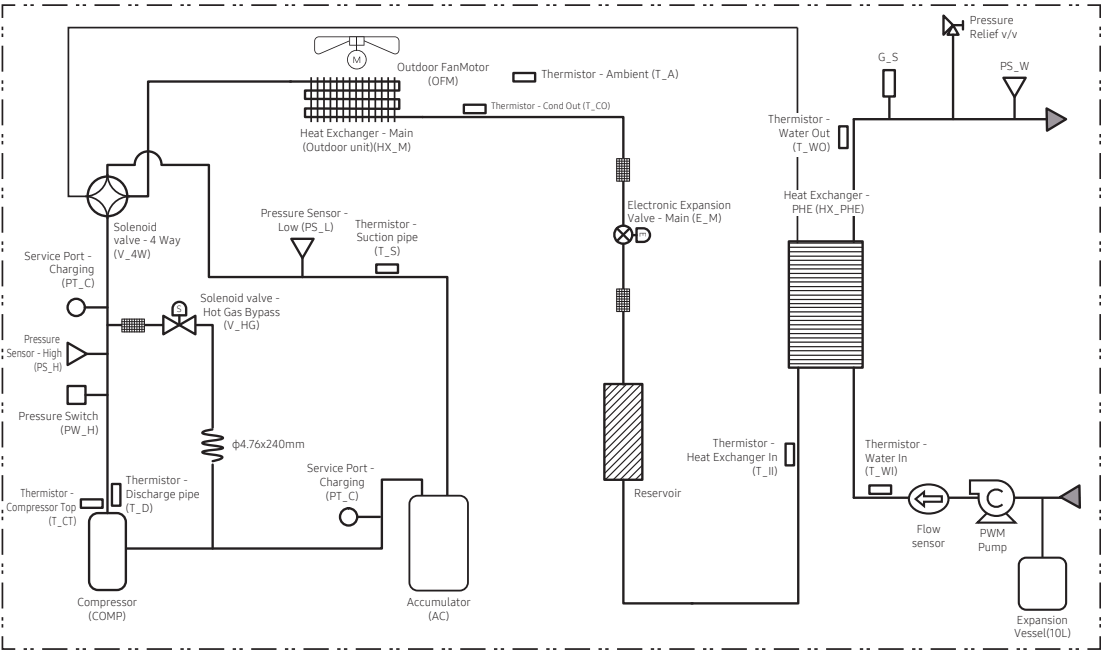


Part	Description
COMP	Compressor
HX_M	Heat Exchanger - Main (Outdoor unit)
HX_PHE	Heat Exchanger - PHE
OFM	Outdoor Fan Motor
AC	Accumulator
PS_H	Pressure Sensor - High
PS_L	Pressure Sensor - Low
PS_W	Pressure Sensor - Water
PW_H	Pressure Switch - High
G_S	Gas Separator
E_M	Electronic Expansion Valve - Main

Part	Description
PT_C	Service Port - Charging
V_4W	Solenoid valve - 4 Way
V_HG	Solenoid valve - Hot Gas Bypass
T_A	Thermistor - Ambient
T_CO	Thermistor - Cond Out
T_CT	Thermistor - Compressor Top
T_D	Thermistor - Discharge pipe
T_II	Thermistor - Heat Exchanger In
T_S	Thermistor - Suction pipe
T_WI	Thermistor - Water In
T_WO	Thermistor - Water Out

Technical data

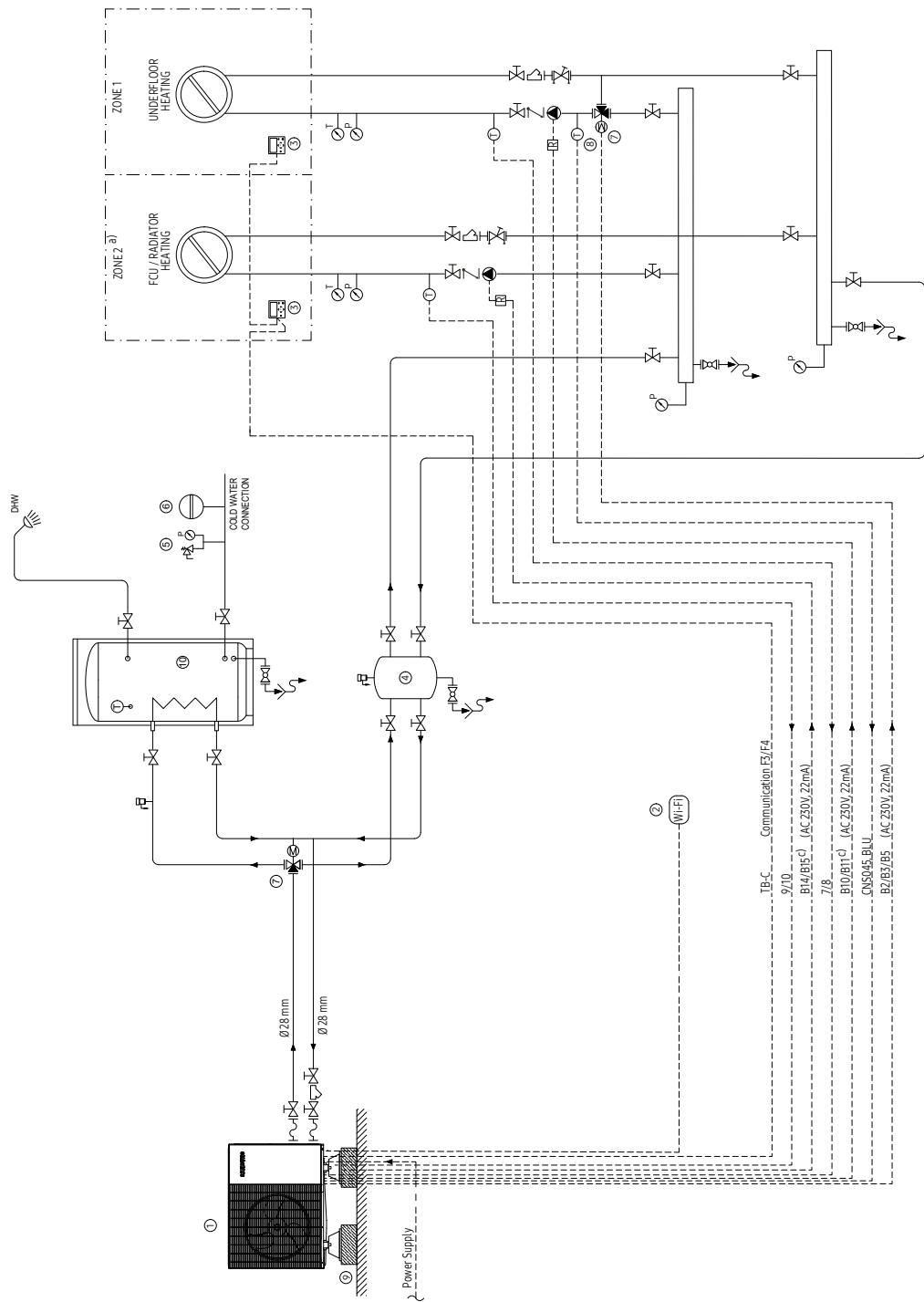
- AE120CXYB**/AE160CXYB**



Part	Description
COMP	Compressor
HX_M	Heat Exchanger - Main (Outdoor unit)
HX_PHE	Heat Exchanger - PHE
OFM	Outdoor Fan Motor
AC	Accumulator
PS_H	Pressure Sensor - High
PS_L	Pressure Sensor - Low
PS_W	Pressure Sensor - Water
PW_H	Pressure Switch - High
G_S	Gas Separator
E_M	Electronic Expansion Valve - Main

Part	Description
PT_C	Service Port - Charging
V_4W	Solenoid valve - 4 Way
V_HG	Solenoid valve - Hot Gas Bypass
T_A	Thermistor - Ambient
T_CO	Thermistor - Cond Out
T_CT	Thermistor - Compressor Top
T_D	Thermistor - Discharge pipe
T_II	Thermistor - Heat Exchanger In
T_S	Thermistor - Suction pipe
T_WI	Thermistor - Water In
T_WO	Thermistor - Water Out

Piping & Wiring diagram



- a) When both zones are simultaneously Thermo On, the operation is performed based on Zone 2. Set the zone with the higher set temperature to Zone 2.
- b) Air vent included in pump integrated outdoor unit. In case that the water piping would be located in a higher position than the air vent of the pump integrated outdoor unit, it is necessary to add additional one at the highest position of the water circuit.
- c) Pumps for radiator and floor heating can be regulated (on/off) by tank integrated hydro unit (terminals B10/B11 and B14/B15) or their own regulators. Based on the current logic only with two zones control with wired remote controller. Not with thermostat.

Technical data

No.	Legend
☐	Samsung EHS pump integrated outdoor unit
☐	Samsung EHS Wi-Fi Kit
☐	Samsung Wired Remote Controller (MWR-WW10N)
☐	Balancing vessel (Decoupler)
☐	Safety group (Safety valve, Manometer)
☐	Expansion vessel
☐	3way mixing valve for floor heating (in case of combination with radiators)
☐	Mixing valve temperature sensor ^{b)}
☐	Vibro-isolating mounting bases
☐	Domestic hot water tank (3rd party)

Symbols	
	Circulation pump
	Shut off valve
	Ball valve
	Check valve
	Strainer
	Temperature gauge
	Pressure gauge
	Safety valve
	3way mixing valve
	Regulation valve
	Temperature sensor
	Air vent (apply when necessary) ^{d)}
	Relay



A3

This appliance is
filled with R-290.